Display mathematics

The Gaussian integral is a classic display:

$$\int_0^1 x^2 \, dx = \frac{1}{3}$$

Fractions and roots grow to fit their contents,

$$\sqrt{\frac{a^2 + b^2}{c^2 + d^2}} \leq \frac{\sqrt{a^2 + b^2}}{\sqrt{c^2 + d^2}} + 1,$$

and a product with limits ends the fixture:

$$\prod_{k=1}^{n} \left(1 + \frac{1}{k}\right) = n + 1.$$

The paragraph after the displays continues the text.